6 Redox Reaction

SOLUTION (QR CODE

2. (B) Let x be the oxidation of Na.
Na has +1 oxidation state and O has –2 oxidation state.

$2(+1) + x + 4(-2) = 0, x = +6$

6. (A) Let x be the oxidation state of N.

N_2O_3:

$2x + 3(-2) = 0, x = +3$

NO:

$x - 2 = 0, x = +2$

NO_2:

$x + 2(-2) = 0, x = +4$

N_2O:

$2x - 2 = 0, x = +1$

So, the correct increasing order of oxidation states of N is:

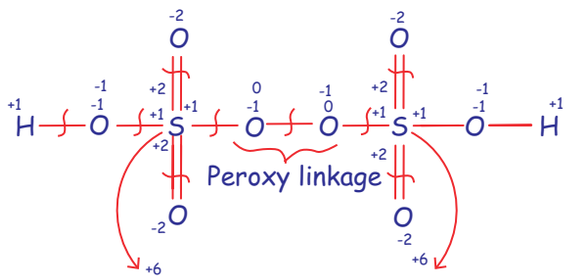
$N_2O > NO > N_2O_3 > NO_2$

agr dimag hai toh itna nahi likhna pdega

9. (B) Let x be the oxidation number of S.
H has +1 oxidation state and –2 is the oxidation state of O.

$2(+1) + 2x + 7(-2) = 0, x = +6$

13.



$2(+1) + 2(x) + 6(-2) + 2(-1) = 0$

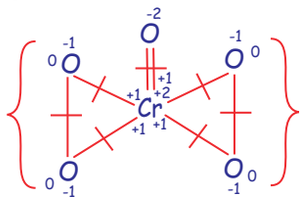
$x = +6$

sara khel babuji structure ka hai

14. $Cr(24) = 1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d^5$

Max O.S = [ns + (n -1)d]

= 1 + 5 = 6



15. (B) Let x be the oxidation state of I.
K has +1 oxidation state and that of O is –2.

$1 + x + 4(-2) = 0, x = +7$

18. (B) Let x be the oxidation state of Fe.
The given complex can be represented as:

$[Fe(CN)_6]^{3-}$

CN has -1 negative charge.

$x - 6 = -3, x = +3$

So, B is the correct option.

20. H_2SO_3, $SOCl_2$, SF_4

Compounds	O.S. of Sulphur
SO_3	+6
H_2SO_3	+4
$SOCl_2$	+4
SF_4	+4
$BaSO_4$	+6
$H_2S_2O_7$	+6

22. (D) Oxidising nature ⇒ O.S. decreases

$$2HI + \overset{+6}{H_2SO_4} \rightarrow I_2 + \overset{+4}{SO_2} + 2H_2O$$

Reduction

are pahle oxidation stage toh nikaalo

29. (C) Oxidation state of sulphur changes from +2 to +2.5, in the reaction of $S_2O_3^{2-}$ with I_2. Oxidation state of sulphur changes from +2 to +6 in the reaction of $S_2O_3^{2-}$ with Br_2.

$\therefore$ Both I_2 and Br_2 are oxidants (oxidising agents) but Br_2 is stronger oxidant than I_2.

thoda sa dimag ko ghumaega

32. (C) In XeF_4, Xe has +4 oxidation state while in XeF_6, it has +6 oxidation state. In O_2F_2,

O has +1 oxidation state while in O_2, it has 0 oxidation state. In this reaction, both reduction and oxidation is taking place simultaneously.

Hence, it is a redox reaction.

34. (B)

List-1 (Reaction)		List-II (Type of Redox Reaction)	
A.	$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$	II.	Combination Reaction
B.	$2NaH \rightarrow 2Na + H_2$	III.	Decomposition Reaction
C.	$V_2O_5 + 5Ca \rightarrow 2V + 5CaO$	IV.	Displacement Reaction
D.	$2H_2O_2 \rightarrow 2H_2O + O_2$	I.	Disproportionation Reaction

kitna asaan hai, hai na, aaya maza

38. Since the compound is reacting with a base so it must be an acid. So n-Factor will be replaceable H^+ ions. It has 'y' H^+ ions so n-factor is 'y'.

thoda toh dimag lagao bacho

41.

$$\overset{+1\,+5\,-2}{AgNO_3} + \overset{+1\,-1}{KI} \longrightarrow \overset{+1\,-1}{AgI} + \overset{+1\,+5\,-2}{KNO_3}$$

$1 + x - 6 = 0$

$x = +5$

Ag^+ NO_3^- K^+ I^-

(n = 1) (n = 1)

जब कुछ नही होता तो

n-factor = total cationic or anionic charge

45. $\underset{\text{Stennous}}{Sn^{2+}} \rightarrow \underset{\text{Stennic}}{Sn^{4+}}$

($KMnO_4$ is strong oxidizing agent, it converts Sn^{+2} into Sn^{+4} and $C_2O_4^{2-}$ to CO_2)

$$\overset{+2}{Sn}\overset{+3}{C_2}O_4 + K\overset{+7}{Mn}O_4 \xrightarrow{H^+} Mn^{2+} + Sn^{4+} + \overset{+4}{C}O_2$$

(n = 5)

2

1 × 2

(n = 4)

4

Reaction toh likho pehle tabhi toh hoga sawaal

48. This is a non-stoichiometric compound in which there exist two oxidation state of iron (+2 and +3).

Applying charge balance equation

Let $Fe^{2+} = x$; $Fe^{+3} = 0.93 - x$

$2(x) + 3(0.93 - x) = 2$

On solving, $x = 0.79$

So in this non-stoichiometric compound

$Fe^{+2} = 0.79$

$Fe^{+3} = 0.93 - 0.79 = 0.14$

$$Fe_{0.93}O \longrightarrow Fe^{+3}$$

$$Fe^{+2}_{0.79} \quad Fe^{+3}_{0.14}$$

$$Fe^{+2}_{0.79} \longrightarrow Fe^{+3}$$

1×0.79

(n = 0.79)

mazedaar sawaal hai, seekhlo

54. Acidic medium.

$$KMnO_4 + SnC_2O_4 \xrightarrow{H^+} Sn^{+4} + CO_2 + Mn^{+2}$$

M·eq of $KMnO_4$ = $\underbrace{\text{molarity} \times \text{n-factor}}_{\text{normality}} \times$ volume

$= 0.2 \times 5 \times V$

M. eq of SnC_2O_4 $= 0.08 \times 4 \times 100$

$= 32$ m. eq

Apply LOE:

$0.2 \times 5 \times V = 32$

$V = 32$ ml.

Samajhne mei Time lagega! Lekin aaega zaroor

55. First calculate weight of chlorine:

Wt of chlorine = 3.1 – 2.3 = 0.8 g

$$\text{Metal} + \overset{0}{Cl_2} \longrightarrow \text{metal}\,\overset{-1}{\text{Chloride}}$$

(n = 2)

LOE

Eq. of metal = Eq of Cl_2

$$\frac{2.3}{E} = \frac{0.8}{\frac{71}{2}}$$

On solving, E = 102.06 g

dekho asaan hai na

59. (C) $Mn_3O_4 \rightarrow MnO_4^- + Mn^{2+}$

$Mn_3O_4 \xrightarrow{n_f = 13} MnO_4^- + 13\,e^-$

$2Mn_3O_4 \xrightarrow{n_f = 2} Mn^{2+}$

$$n_{eff} = \frac{2\times13}{2+13} = \frac{26}{15}$$

$$E = \frac{M}{26} \times 15 = \frac{15M}{26}$$

n-factor aaata hai ya nahi, lallu

60. (C) $\text{metal} + \overset{0}{O_2} \rightarrow \text{metal}\,\overset{-2}{\text{oxide}}$

2 × 2 = 4

(n = 4)

n-factor of oxygen is 4 so its equivalent wt = $\frac{32}{4} = 8$

According to LOE:

Eq. of metal = Eq of O_2

Let the weight of metal = x_g

weight of $O_2 = \frac{24x}{100}g$

$$\frac{x}{E} = \frac{\left(\frac{24x}{100}\right)}{8}$$

wt, wt, Eq wt, Eq wt

E = 33.3

ghabrao mat, itna bh mushkil nahi hai

63. (C) Number of meq of $Na_2CO_3.xH_2O$ in 20 ml

$$= 19.8 \times \frac{1}{10} = 1.98$$

∴ Number of milliequivalents of $Na_2CO_3.xH_2O$ in 100 mL

$$= 1.98 \times \frac{100}{20} = 9.9$$

$$\frac{\text{Mass}}{\text{Equivalent mass}} \times 1000 = 9.9$$

$$\frac{0.7}{\frac{M}{2}} \times 1000 = 9.9,\ M = 141.40$$

106 + 18x = 141.4

On solving, x = 2

dimag mat ghumao

64. (B) 200 mL (0.2 L) of KI solution (0.1 M) → Moles of KI (I^-)

= 0.1 × 0.2 = 0.02 moles.

From reaction: 5 moles of I^- react with 1 mole of IO_3^-.

Required IO_3^-: 0.02 × 1/5 = 0.004 moles.

(d) is correct

Equivalent weight formula:

$$\text{Equivalent weight} = \frac{\text{Molecular weight}}{\text{n-factor}}$$

For IO_3^- in this reaction, the change in oxidation state (n-factor) is 5.

Hence, Equivalent weight = (Molecular weight)/5.

isko dekhlo dhyan se

68. $1NaOH + 1HCl \rightarrow NaCl + H_2O$

83 × 0.45 mmol 235 × M

Mole concept **(According to balanced chemical equation)**

83 × 0.45 = 235 × M

$$M = \frac{83\times0.45}{235} = 0.16\ M$$

Equivalent concept

M. eq of HCl = M. eq of NaOH

235 × M × 1 = 83 × 0.45 × 1

$$M = \frac{83\times0.45}{235} = 0.16\,M$$

dono hi treeeke lazawaab ha

75. Reaction of $Ca(OH)_2$ with HCl is

$Ca(OH)_2 + 2HCl \rightarrow CaCl_2 + 2H_2O$

$$M_{Ca(OH)_2} = \frac{\text{no. of millimoles}}{V(ml)} = \frac{5}{10}$$

Reaction of $Ca(OH)_2$ with H_2SO_4 is

$Ca(OH)_2 + H_2SO_4 \rightarrow +CaSO_4 + 2H_2O.$

No. of millimoles of $Ca(OH)_2$

$= 20 \times 0.5 = 10$

i.e., no. of millimoles of $H_2SO_4 = 10$

$$\Rightarrow M_{H_2SO_4} = \frac{\text{no. of millimoles}}{V(ml)} = \frac{10}{10} = 1\ M$$

Hence, concentration of $H_2SO_4 = 1M$

reaction toh aati hai na likhni

79. $Cu^{+2} + I^- \rightarrow I_2 + Cu^+$ (n = 1)

$$Cu^{2+} + \overset{+3}{C_2}O_4^{2-} \xrightarrow[H^+]{KMnO_4} Cu^{+2} + \overset{+4}{C}O_2 \quad (1 \times 2)$$

Eq of $C_2O_4^{2-}$ = Eq of $KMnO_4$

$$\text{Eq of } C_2O_4^{2-} = \frac{22.6 \times 0.02 \times 5}{1000}$$

$$\text{mol of } C_2O_4^{2-} = \frac{22.6 \times 0.02 \times 5}{1000 \times 2}$$

Eq of $Na_2S_2O_3$ = Eq of I_2

= Eq of I^- = Eq of Cu^{+2}

$$\text{Eq of } Cu^{+2} = \frac{11.3 \times 0.05 \times 1}{1000}$$

$$I_2 + Na_2\overset{2}{S_2}O_3 \rightarrow Na\overset{-1}{I} + Na_2\overset{2.5}{S_4}O_6$$

(n = 2) (n = 1)

$$\text{Moles of } Cu^{2+} = \frac{11.3 \times 0.05 \times 1}{1000 \times 1}$$

$$\frac{\text{Moles of } Cu^{2+}}{\text{Moles of } C_2O_4^{2-}} = \frac{\dfrac{11.3 \times \cancel{0.05} \times 1}{\cancel{1000}}}{\dfrac{22.6 \times \cancel{0.05} \times \cancel{2}}{\cancel{1000} \times \cancel{2}}}$$

$$= \frac{11.3}{22 \cdot 6} = \frac{1}{2}$$

tedha sawaal hai

80. $2KClO_3 + 12HCl \rightarrow 2KCl + 6H_2O + 6Cl_2$

$Cl_2 + 2KI \rightarrow 2KCl + I_2$

Also meq. of I_2 = meq. of hypo = 100 × 1

$\therefore [N_{Hypo} \times 12.3 = 24.6 \times 0.5, \therefore N_{Hypo} = 1]$

$$\text{Also mM of } Cl_2 = \text{mM of } I_2 = \frac{100}{2} = 50$$

$$\text{Also mM of } KClO_3 = \frac{2 \times \text{mM of } Cl_2}{6} = \frac{2 \times 50}{6} = \frac{50}{3}$$

$$\therefore \frac{w_{KClO_3}}{122.5} \times 1000 = \frac{50}{3} \Rightarrow w_{KClO_3} = 2.042$$

$$\therefore \ \%KClO_3 = \frac{2.042}{2.48} \times 100 = 82.34\%$$

agr sawaal ki bhasha samjh gye toh matlab kr gaye isko

83. $KMnO_4 + K_2Cr_2O_7 + K\overset{-1}{I} \rightarrow \overset{0}{I_2}$

(n = 5) (n = 6)

$$I_2 + Na_2\overset{2}{S_2}O_3 \longrightarrow NaI + Na_2\overset{2.5}{S_4}O_6 \quad \text{(learn)}$$

(Hypo Solution)

Eq of $KMnO_4$ + Eq of $K_2Cr_2O_7$ = Eq of KI

Eq of I_2 = Eq of Hypo sol^n.

Let x be the mass of $K_2Cr_2O_7$ and (0.561 – x)g be the mass of $KMnO_4$.

$$\frac{x}{M_{K_2Cr_2O_7}} \times 6 + \frac{0.561 - x}{M_{KMnO_4}} \times 5 = \frac{100 \times 0.15 \times 1}{1000}$$

948x + 824.67 – 1470x = 696.78

–522x = –127.89

x = 0.245, 0.561 – x = 0.316

$$\%K_2Cr_2O_7 = \frac{0.245}{0.245 + 0.316} \times 100\% = 43.6\%$$

% $KMnO_4$ = (100 – 43.6) = 56.4%

Jo bhidte hain unke barabar hote hain

88. Total oleum = 100 g → 40 g SO_3, 60 g H_2SO_4

$SO_3 + H_2O \rightarrow H_2SO_4$

$$\frac{40}{80}\text{mol} \rightarrow \frac{40}{80}\text{mol} \rightarrow \frac{40}{80} \times 98\text{gm}$$

New H_2SO_4 + Purana H_2SO_4 = labelling

$\frac{40}{80} \times 98 + 60 = 109\,gm$

So, the labelling is 109%.

90. First determine the oxidation states of the species involved.

$$\overset{+2}{Fe^{2+}} + \overset{+6}{Cr_2}O_7^{2-} \rightarrow \overset{+3}{Fe^{3+}} + \overset{+3}{Cr^{3+}}$$

Separate the reaction into two halves: Oxidation and reduction followed by balancing of atoms:

Oxidation half $Fe^{2+} \rightarrow Fe^{3+} + e^- \times 6$

Reduction Half $Cr_2O_7^{2-} \rightarrow 2Cr^{+3}$

$6e^- +$

$$6Fe^{2+} \rightarrow 6Fe^{3+} + 6e^-$$

$$14H^+ + 6e^- + Cr_2O_7^{2-} \rightarrow 2Cr^{+3} + 7H_2O$$

Net reaction is:

$$6Fe^{2+} + Cr_2O_7^{2-} + 14H^+ \rightarrow 6Fe^{3+} + 2Cr^{3+} + 7H_2O$$

mazedaar hai yeh sawaal

93. (A)

$$2KMnO_4 + 5H_2\overset{-2}{S} + 6H^+ \rightarrow 2\underset{\text{Oxidation}}{\overset{+2}{Mn^{2+}}} + 2K^+ + 5\overset{0}{S} + 8H_2O$$

$H_2S \Rightarrow R.A \Rightarrow$ Reductant

So, $\overset{-2}{S} \rightarrow \overset{0}{S} + 2e^-$

2 moles of electrons are involved in oxidation of 1 mole of reluctant.

Therefore, option (A) is the correct answer.

balance toh ab kar hi loge

94. (B) $Fe + HNO_3 \rightarrow Fe(NO_3)_2 + NH_4NO_3 + H_2O$

Half reaction method:

$Fe \rightarrow Fe^{2+} + 2e^-$ ...(1) × 4

$NO_3^- + 10H^{\oplus} + 8e^{\ominus} \rightarrow NH_4^{\oplus} + 3H_2O$ ...(2)

Equation (1) × 4 + (2)

$4Fe + NO_3^- + 10H^{\oplus} \rightarrow 4Fe^{2+} + NH_4^{\oplus} + 3H_2O$

Add^n of spectator ion

$4Fe + 10HNO_3 \rightarrow 4Fe(NO_3)_2 + NH_4NO_3 + 3H_2O$

Therefore, option (B) is the correct answer.